

TEAM SONAR

REEFPRINT (developed as KHANYA) — Mintek SCI Grad Hackathon 2026

23 August 2026

Boitumelo Lekalakala
Mintek SCI Grad Hackathon 2026
Mintek
Private Bag X3015, Randburg 2125

Request for a Mintek-assigned mentor — Team Sonar

Dear Boitumelo Lekalakala,

Your letter of 14 August 2026 asks each team either to supply a mentor's name and contact details or to indicate clearly if a mentor from Mintek is required. **Team Sonar does not have a mentor, and we are requesting that Mintek assign one.**

We are three students at three institutions: Lethabo Hoaeane (BCom Business Informatics, University of South Africa), Sibusiso Khumalo (BSc Electrical Engineering, University of the Witwatersrand) and Ipeleng Modise (BSc Computer Science, Tshwane University of Technology). We share no campus and no common supervisor, which is why we have no mentor to name.

Our entry, REEFPRINT (developed as KHANYA), addresses the *Computer Vision for Real-Time Mineralogical Characterisation* challenge: automated identification of opaque ore minerals in polished sections of UG2 ore, by recovering the full linear Stokes vector at every pixel from a rotating-analyser image series. It is software, evaluated on public data; no instrument is built.

So that you can match us appropriately, the areas in which mentorship would most improve the work, in order of value to us — any one of them would be valuable:

1. **Reflected-light ore microscopy and process mineralogy** — quantitative reflectance, bireflectance and anisotropy on polished sections. None of us is a practising mineralogist, and several of our load-bearing claims are mineralogical.
2. **UG2 and Merensky flotation, and PGE deportment** — to test whether the three operator-facing outputs we have chosen (fine-chromite entrainment risk, naturally-floating-gangue load for depressant dosing, and a stockpile oxidation index) are the ones a concentrator would act on.
3. **Automated mineralogy in practice (QEMSCAN, MLA)** — to check our characterisation of what SEM-based automated mineralogy does and does not deliver inside a control loop.

We are not requesting data, samples or laboratory access: we work under a self-imposed rule of public sources only, and no proprietary Mintek data is used anywhere in this project. What we are asking for is expert review — someone willing to tell us where we are wrong about the mineralogy, early enough that it changes what we build.

We appreciate that mentor time is scarce. Two or three short conversations between now and 1 October would materially improve the work. We will prepare questions in advance, keep to time, and fit whatever schedule and format suit the mentor; remote is easiest for us, and we can meet outside working hours.

Thank you for the opportunity, and for considering this request.

Yours sincerely,

Lethabo Hoaeane

Team Sonar — technical lead and point of contact
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On behalf of Sibusiso Khumalo and Ipeleng Modise.